

4.2 $p(x) = a_4 x^4 + a_3 x^3 + a_2 x^2 + a_1 x + a_0$, $a_0, a_1, a_2, a_3, a_4 \in \mathbb{R}$

$$p(0) = 1 = a_0$$

$$p(1) = 2 = a_0 + a_1 + a_2 + a_3 + a_4$$

$$p(2) = -3 = a_0 + 2a_1 + 4a_2 + 8a_3 + 16a_4$$

$$\left(\begin{array}{cccc|c} a_1 & a_2 & a_3 & a_4 & \\ 1 & 1 & 1 & 1 & 1 \\ 2 & 4 & 8 & 16 & -4 \end{array} \right) \sim \left(\begin{array}{cccc|c} 1 & 2 & 4 & 8 & -2 \\ 0 & 1 & 3 & 7 & -3 \end{array} \right) \quad \begin{array}{l} a_1 = -2 + 6s + 6 + 14t - 4s - 8t = 4 + 2s + 6t \\ a_3 = s \quad a_4 = t \quad a_2 = -3 - 3s - 7t \end{array}$$

$$\left\{ \begin{pmatrix} 4 \\ -3 \\ 0 \\ 0 \end{pmatrix} + s \begin{pmatrix} 2 \\ -3 \\ 1 \\ 0 \end{pmatrix} + t \begin{pmatrix} 6 \\ -7 \\ 0 \\ 1 \end{pmatrix} \right\}$$

$$\|\vec{u}\|^2 = a_0^2 + a_1^2 + a_2^2 + a_3^2 + a_4^2 \quad \text{co nejmenší}$$

$$\vec{v} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{pmatrix}$$

$$\text{také co nejmenší} \wedge \vec{v} \vec{v}_1 = 1, \vec{v} \vec{v}_2 = -2$$

$$a_1 = 1\alpha + 1\beta$$

$$a_2 = 1\alpha + 2\beta$$

$$a_3 = 1\alpha + 4\beta$$

$$a_4 = 1\alpha + 8\beta$$

$$\left. \begin{array}{l} a_1 = 1\alpha + 1\beta \\ a_2 = 1\alpha + 2\beta \\ a_3 = 1\alpha + 4\beta \\ a_4 = 1\alpha + 8\beta \end{array} \right\} \text{dosazujeme } \alpha\text{-násobek vektoru } \vec{v}_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} \text{ a } \beta\text{-násobek } \vec{v}_2 = \begin{pmatrix} 1 \\ 2 \\ 4 \\ 8 \end{pmatrix}$$

$$4\alpha + 15\beta = 1$$

$$\alpha + \beta + 2(\alpha + 2\beta) + 4(\alpha + 4\beta) + 8(\alpha + 8\beta) = 15\alpha + 85\beta = -2$$

$$\left(\begin{array}{cc|c} 4 & 15 & 1 \\ 15 & 85 & -2 \end{array} \right) \sim \left(\begin{array}{cc|c} 4 & 15 & 1 \\ 0 & 115 & -23 \end{array} \right) \quad \begin{array}{l} \alpha = \frac{1}{4} + \frac{15}{115} = \frac{5+15}{20} = 1 \\ \beta = -\frac{23}{115} = -\frac{1}{5} \end{array}$$

$$a_1 = 1\alpha + 1\beta = 1 - \frac{1}{5} = \frac{4}{5}$$

$$a_2 = 1\alpha + 2\beta = 1 - \frac{2}{5} = \frac{3}{5}$$

$$a_3 = 1\alpha + 4\beta = 1 - \frac{4}{5} = \frac{1}{5}$$

$$a_4 = 1\alpha + 8\beta = 1 - \frac{8}{5} = -\frac{3}{5}$$

$$\underline{\underline{p(x) = -\frac{3}{5}x^4 + \frac{1}{5}x^3 + \frac{3}{5}x^2 + \frac{4}{5}x + 1}}$$